

Project Proposal

MOBILE ATTENDANCE SYSTEM

Course: Information Security

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Project Focus	Mobile Attendance with location-based attendance verification

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BSCS 3 — Section B

Table of Contents

1. Introduction
 2. Problem Statement
 3. Proposed Solution
 4. Project Objectives
 5. System Flowchart
 6. User Roles and Permissions
 7. Main Features
 8. Information Security Features
 9. Technology Stack
 10. Expected Outcome
 11. Conclusion
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1. Introduction

Attendance management is essential in educational institutes and organizations. Traditional attendance methods often lead to proxy attendance, inaccurate records, and security concerns. This project proposes a Mobile Attendance System that uses GPS-based location verification to ensure users can mark attendance only from authorized locations.

2. Problem Statement

Current attendance systems face several challenges:

- Proxy attendance and fake check-ins.
 - Attendance marking from unauthorized locations.
 - Inaccurate attendance records.
 - Lack of proper monitoring and reporting.
 - Weak access control and security mechanisms.
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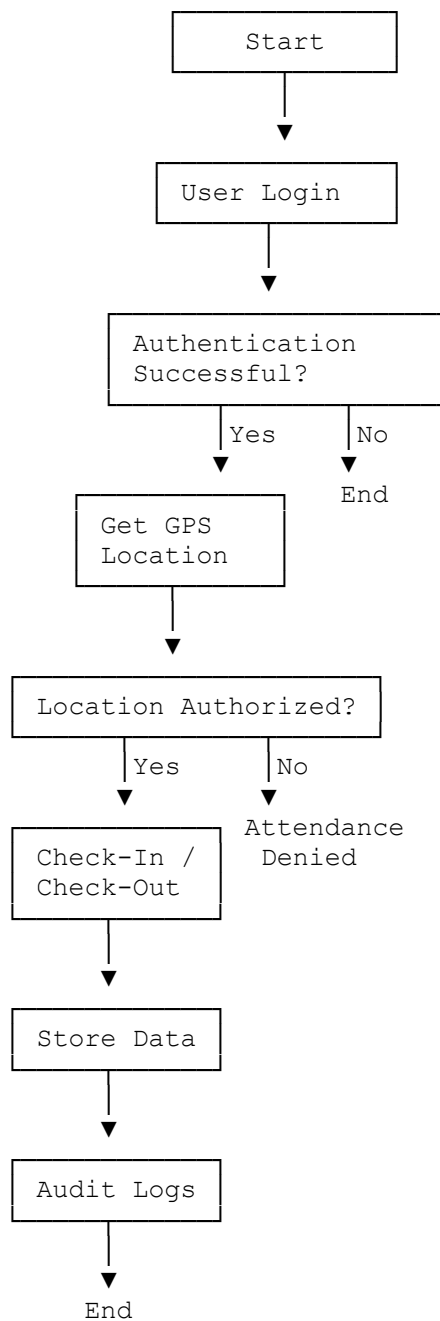
3. Proposed Solution

The proposed Mobile Attendance System allows users to check in and check out through a mobile application. The system verifies the user's GPS location before enabling attendance actions. Role-based access control, audit logs, and attendance reports help ensure secure and reliable attendance management.

4. Project Objectives

- Develop a mobile-based attendance system.
 - Implement GPS-based location verification.
 - Prevent unauthorized attendance marking.
 - Provide role-based access control.
 - Generate attendance reports.
 - Maintain secure audit logs.
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5. System Flowchart



6. User Roles and Permissions

Role	Permissions
Admin	Manage users, locations, attendance records, reports, and settings
Teacher / Manager	View attendance records and generate reports
Student / Employee	Check-In, Check-Out, and view attendance history

7. Main Features

Feature	Description
Secure Login	User authentication and authorization
GPS Verification	Verifies user location before attendance
Check-In / Check-Out	Attendance recording with timestamps
Role-Based Access Control	Different permissions for different roles
Attendance Reports	Daily and monthly reports
Audit Logs	Tracks important activities
GPS Spoofing Detection	Detects fake GPS attempts
Biometric Verification	Fingerprint or Face ID authentication

8. Information Security Features

Security Concept	Implementation
Authentication	Secure login using JWT
Authorization	Role-based permissions
Data Integrity	Protected attendance records
Confidentiality	Restricted data access
Audit Logging	Activity tracking and monitoring
Fraud Prevention	GPS verification and spoofing detection

9. Technology Stack

Layer	Technology
Mobile Application	Flutter
Backend	Node.js + Express.js
Database	PostgreSQL / Supabase
Authentication	JWT
Location Services	Geolocator / Google Maps API
Admin Dashboard	React.js
Reports	PDF / CSV Generation

10. Expected Outcome

The system will provide a secure and reliable attendance management solution by ensuring attendance can only be marked from authorized locations. It will improve attendance accuracy, reduce proxy attendance, and strengthen security through authentication, authorization, and location verification.

11. Conclusion

The Mobile Attendance System combines location-based attendance verification with essential Information Security concepts such as authentication, authorization, role-based access control, audit logging, and fraud prevention. The project offers a practical, secure, and scalable solution for attendance management in educational institutes and organizations.